



Adnan Abbas Shah

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PROFESSIONAL PROFILE

Remote Sensing and GIS Expert specializing in machine learning, cloud-based Earth observation, and multi-sensor satellite analytics for agriculture, climate, disaster risk, and environmental monitoring. Delivered 29+ national and international geospatial projects and supported MS and PhD-level research workflows from spatial data processing and modeling to publication-ready maps, dashboards, and environmental decision-support outputs.

RESEARCH INTERESTS

- Remote Sensing and GIS for Environmental Monitoring
- Geospatial Data Science and Machine Learning
- Climate Change, Drought, and Hydroclimatic Extremes
- Disaster Risk Assessment and Spatial Modeling
- Agricultural Land Dynamics and Food Security
- Cloud Computing (GEE) and Earth Observation Analytics
- Water Resource Management

EDUCATION

- **COMSATS University Islamabad** Feb 2023 – Mar 2025
Master of Science in Remote Sensing and GIS Islamabad, Pakistan
 - CGPA: 3.62/4.00 Advisor: Prof. Dr Muhammad Farooq Iqbal
 - Major Courses: Geostatistics, Satellite Meteorology, Advanced Remote Sensing, Advanced GIS, Land Surveying
 - Research: *ML-Based Agricultural Land Degradation Assessment due to Rapid Urbanization* – Quantified agricultural land conversions (1990–2023) across Punjab, Pakistan. Applied ML/DL for soil salinity classification. Assessed vegetation trends in desert regions using ML.
- **Government College University Faisalabad** Mar 2018 – Nov 2022
Bachelor of Science in Geography Faisalabad, Pakistan
 - CGPA: 3.81/4.00
 - Major Courses: Physical Geography, Human Geography, Economic Geography, GIS, Remote Sensing, Environmental Science, Statistics, Computer Sciences, Research Methodology, Climate Change

PUBLICATIONS

C=CONFERENCE, J=JOURNAL

- [C.1] **Shah, A.A., & Iqbal, M.F. (2024). Flood Susceptibility Mapping in District Jhang Using Remote Sensing and Advanced Machine Learning Techniques.** In *2024 International Conference on Frontiers of Information Technology (FIT)*, pp. 1–6. IEEE.
- [J.1] Nawaz, M., Kalisa, W., Zaheen, Z., Tauqir, M., Zhang, J., **Shah, A.A.**, & Ullah, K. (2026). **Spatiotemporal Assessment of Desertification Sensitivity in Ningxia, China, Using the MEDALUS Framework and Random Forest Classification (2001–2022).** *Journal of Geoscience and Earth Observation*, 1(1), 39–54.
- [J.2] **Shah, A.A.** (sole author, 2026). **Assessing the Impacts of Urbanization and Climate Variability on Cropland Dynamics and Food Security in Punjab, Pakistan (2000–2025): A Multi-Scale Remote Sensing and Machine Learning Approach.** *Environmental Monitoring and Assessment*, 198(7), 700. Springer.
- [J.3] Ali, O., Ahmad, J., **Shah, A.A.**, & Sohail, M. (2026). **Hydrometeorological Assessment of Water Budget Components in Rawalpindi Division, Pakistan: An ERA5 Reanalysis and Machine Learning Approach.** *Journal of Nautical Eye and Strategic Studies*, 6(1), 87–107.

MANUSCRIPTS UNDER REVIEW

- [S.1] Shah, A.A. (sole author). **Entropy-Based Characterization of Drought and Climate Change Amplification in the Jhelum River Basin, Pakistan (1981–2024)**. Submitted to *Hydrological Sciences Journal*, Taylor & Francis.
- [S.2] Shah, A.A. (sole author). **The Retreating Frontier: Quantifying the Accelerating Poleward Shift of the Atlantification Boundary in the Barents Sea**. Submitted to *Environmental Monitoring and Assessment*, Springer.
- [S.3] Shah, A.A. (corresponding author), Abbas, A., Khalid, M., & Syed, J.H. **Compound Drought and Heatwave Events over Pakistan (1981–2025): Observed Trends, Coupling Mechanisms, and Future Projections under CMIP6 Scenarios**. Submitted to *Natural Hazards*, Springer.
- [S.4] Abid, H., Iqbal, J., Shah, A.A. (corresponding author), & Zafar, A. **Analyzing Water Demand and Supply under Socioeconomic and Climate Change Scenarios using WEAP: A Case Study of the Kabul River Basin, Pakistan**. Submitted to *Water Conservation Science and Engineering*, Springer.
- [S.5] Khan, I., Iqbal, J., Shah, A.A. (corresponding author), et al. **Multi-Decadal Shoreline Dynamics of the Gwadar Coast, Pakistan: Insights from DSAS and Satellite Remote Sensing Data**. Submitted to *Modeling Earth Systems and Environment*, Springer.
- [S.6] Saleem, H., Iqbal, J., & Shah, A.A. (corresponding author). **Adding Climate Data Improves Wildfire Susceptibility Prediction More Than Model Complexity: A Controlled Comparison in Abbottabad District, Pakistan**. Submitted to *Natural Hazards*, Springer.
- [S.7] Shah, A.A. (sole author). **Marine Heatwaves and Cold Spells in the Tropical and North Indian Ocean (1982–2025): Trends and Climate-Mode Drivers**. Submitted to *Earth System Dynamics*, Copernicus/EGU. Preprint: EGU sphere (egusphere-2026-2971).

EXPERIENCE

- **Sustainable Environmental Management (SEM) Pvt. Ltd.** May 2025 – Present
GIS Consultant Islamabad, Pakistan
 - Led geospatial project delivery, environmental impact assessment support, and client-facing advisory on remote sensing-based solutions
 - Oversaw Web GIS dashboards, spatial database/server management, and API integration for consultancy projects
 - Directed predictive modeling, time series analysis, and spatiotemporal anomaly detection workflows
- **Sustainable Environmental Management (SEM) Pvt. Ltd.** May 2024 – Apr 2025
GIS Specialist Islamabad, Pakistan
 - Performed predictive modeling, time series analysis, and spatiotemporal anomaly detection
 - Developed Web GIS dashboards and spatial applications using GEE, Python, and JavaScript
 - Supported cadastral mapping, API integration, and geospatial database/server management
- **Sustainable Environmental Management (SEM) Pvt. Ltd.** May 2023 – Apr 2024
GIS Analyst Islamabad, Pakistan
 - Conducted spatial data processing, visualization, and map generation for environmental and geospatial projects
 - Supported remote sensing-based analysis and environmental impact assessment tasks
 - Assisted in geospatial database management and preparation of publication-ready maps
- **Environmental & Air Quality Research Lab, COMSATS University Islamabad** Nov 2022 – Apr 2023
GIS Analyst (Advisor: Dr. Jabir Hussain Syed) Islamabad, Pakistan
 - Supported geospatial analysis for air quality monitoring, modeling, and prediction using GIS tools
 - Collaborated on organizing a workshop on Plastic Waste Recycling awareness and research
 - Conducted spatial data processing, visualization, and map generation for environmental research projects

PRESENTATIONS

- **Poster: ML-Based Agricultural Land Degradation Assessment** 2025
Career Expo 2025, COMSATS University, Islamabad
- **Poster: Temporal LULC Change Analysis of District Jhang** 2022
GIS Exhibition 2022, Government Graduate College, Jhang

GEOSPATIAL TOOLS & TECHNICAL SKILLS

- **Cloud & Geospatial Computing:** Google Earth Engine, Google Colab, Kaggle
- **GIS & Remote Sensing Platforms:** ArcGIS Pro, ArcMap, QGIS, SNAP, ENVI, ERDAS Imagine, TerrSet/IDRISI, HEC-RAS
- **Programming & Geospatial Data Science:** Python, R, JavaScript, SQL, GeoPandas, Rasterio, Xarray, GDAL, Pandas, NumPy
- **Remote Sensing & Geospatial Modeling:** Multi-sensor Earth observation analytics, data fusion, spatiotemporal modeling, geospatial simulation, environmental change assessment, climate-risk mapping, Landsat, Sentinel, MODIS, TROPOMI, ERA5, and CHIRPS
- **Machine Learning & GeoAI:** Classification, regression, data augmentation, anomaly detection, spatial prediction, image segmentation, feature engineering, data fusion, and spatial downscaling.
- **Research & Scientific Communication:** Methodology development, academic report writing, manuscript preparation, peer-review response writing, technical documentation, publication-ready maps, scientific figures, research presentations

HONORS AND AWARDS

- **Best Student of the Session 2018–2022** 2022
Government College University Faisalabad
 - Awarded in recognition of academic excellence and outstanding performance
- **PEEF Scholarship for Academic Achievement**
Punjab Educational Endowment Fund
 - Fully funded scholarship awarded for academic excellence
- **PM Laptop Scheme Scholarship** 2023–2024

SELECTED PROJECTS

- **Flood Susceptibility Mapping: AHP vs. ML for EWS**
Tools: Python, GEE, Random Forest, XGBoost, AHP
 - Comparative analysis of Analytical Hierarchy Process and ML methods for early warning systems in DG Khan
- **Cloud-Based Flood Susceptibility Web Application**
Tools: Google Earth Engine, JavaScript, ML algorithms
 - Developed a cloud-based web app for real-time spatial analysis and flood risk visualization on GEE
- **High-Resolution Climate Dataset for South Asia**
Tools: Google Earth Engine, Python, MODIS, ML
 - Gap-filled and downscaled MODIS and bioclimatic variables using ML and cloud computing
- **2025 Flood Mapping and Damage Assessment, Punjab** 2025
Tools: Satellite imagery, GIS
 - Temporal flood mapping and damage assessment using satellite imagery and GIS-based analysis
- **Avalanche Susceptibility Mapping, Gilgit-Baltistan**
Tools: ML algorithms, Remote Sensing
 - Applied ML algorithms and RS data for hazard risk assessment in mountainous terrain

CERTIFICATIONS

- **Penn State University (Coursera):** Maps and the Geospatial Revolution
- **Penn State University (Coursera):** Introduction to GIS Mapping
- **Alison (NPTEL/IIT):** Advanced Diploma in Geographic Information Systems
- **DigiSkills.pk:** WordPress Development

LANGUAGE SKILLS

Mother tongue(s): Urdu | Punjabi

🇬🇧 English

👂 LISTENING: C2 📖 READING: C2 ✍️ WRITING: C1

🗣️ SPOKEN PRODUCTION: C1

🗣️ SPOKEN INTERACTION: C1

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user

REFERENCES

1. **Dr. Kalim Ullah**
Chairman / Tenured Professor of Meteorology
COMSATS University Islamabad
Email: kalim_ullah@comsats.edu.pk
Relationship: Department Chairman
2. **Dr. Jabir Hussain Syed**
Tenured Associate Professor, China Study Center
COMSATS University Islamabad
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Relationship: Research Lab Advisor
3. **Dr. M. Farooq Iqbal Buzdar**
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COMSATS University Islamabad
Email: farooq_iqbal@comsats.edu.pk
Relationship: MS Thesis Advisor